

Syllabus for System Dynamics, MCEN 4043, Fall 2002

- INSTRUCTOR: Michael Sprague, ECME 219, 492-0229, Michael.Sprague@Colorado.edu
Office Hours: Mon 1:00 - 2:00, Wed 12:00 - 1:00, ECME 215A
- TAs: Grader: Ming Yi, ECME 219, Ming.Yi@Colorado.edu
For questions on grading, please e-mail Ming for an appointment.
Recitation: Dhruv Bhate, ECME 209, Dhruv.Bhate@Colorado.edu
Office Hours: Tue 5:00 - 6:00, ECME 209
- PREREQUISITES: MCEN 3017 (Circuits & Electronics), MCEN 3021 (Fluid Mechanics), MCEN 3022 (Heat Transfer), and MCEN 3043 (Dynamics)
- TEXT: K. Ogata, *System Dynamics*, 3rd edition
- OUTLINE: Mechanical Systems (Ch. 3)
Electrical Systems (Ch. 4)
Complex Variables & Laplace Transforms (Ch. 2)
Analysis in the Time Domain (Ch. 6)
Analysis in the Frequency Domain (Ch. 7)
Analysis & Design in the Time Domain (Ch. 8)
Analysis & Design in the Frequency Domain (Ch. 9)
- HOMEWORK: Given each Thursday and due at the *beginning* of class on the following Thursday. Homework submitted by 5 PM on the next day will receive half credit (put in Mike's mailbox); later homework will not be credited. Homework solutions will be placed in a folder in the Engineering Library (downstairs, by the copy machines) and on the board across from ECME 269. Please make copies of these solutions and return the originals to their proper location (stolen copies of the homework sets will not be replaced). Staple papers in the upper left corner, do not fold, and do not use paper from a spiral notebook; engineering paper is preferred. Working with others is fine, but **don't copy someone else's solution!** Copying previously posted solutions will be considered cheating.
- QUIZZES: In-class pop quizzes (5-10 min.) will be given occasionally. For preparation, study the notes of the previous lecture.
- EXAMS: There will be two closed-book midterms and one final. Regarding "crib sheets," you are allowed *one side* of an 8.5 in. × 11 in. sheet for Exam 1, *two sides* for Exam 2 and *two sheets* for the final.
- GRADING:
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| Homework | 20% | (lowest score dropped) |
| Quizzes | 10% | (lowest score dropped) |
| Exam 1 | 20% | Tuesday, October 1 |
| Exam 2 | 20% | Thursday, November 7 |
| Final | 30% | Saturday, Dec. 14, 7:30 PM - 10:00 PM |

- RECITATION: Monday 4:00 - 5:00, ECME 215A. Dhruv will present example problems that parallel the week's homework assignment. Please do not use the recitation to ask specific homework questions; save those for office hours.
- E-MAIL LIST: Please sign up for the class e-mail list *immediately*. To do this, send a message to listproc@lists with the subject blank, and the following message body: subscribe mcen4043-f02 <your full name>.
- COMPUTER USE: I highly recommend using software like MatLab and Mathematica to help with the homework solutions; it's an easy way to check your answers. Also, system-response plots can provide useful insight. Finally, use your judgment when using software, *i.e.*, show enough of *your* work when turning in an assignment.